Pharmakon — work split

Anish Ganapa & Punya Gandhi · drafted 6 September 2026 · target: end of September

Scope decision, made once

The Overview document describes four things: a PK safety engine, a correlation/pattern engine, plain-English symptom matching, and a caregiver dashboard. The Technical Specification builds only the first. The Technical Specification is what we build.

With two people and roughly twenty-four days, we can build the PK engine well or all four badly. The pattern engine alone — multiple-comparisons correction, lagged effects, staying quiet when the data does not support a finding — is its own month of work, and its failure mode is confidently telling someone that dairy causes their acne when it is noise. That is worse than not shipping it.

The rule that makes a two-person split work

The API contract is the seam, not backend-versus-frontend. `docs/api-contract.md` is frozen at the end of week 1 and changes afterward only by agreement of both people, in a pull request that touches nothing else.

Punya builds against a mock server from day one and never blocks on the engine being finished. Anish builds the engine against tests, not against the UI. Neither person's daily progress depends on the other's.

Anish — engine, API, validation

Owns app/pk/, app/detect/, app/api/, the substance registry, and the FAERS backtest.

- [x] Bateman single dose, superposition, Cockcroft–Gault — `app/pk/`
- [x] Duplication and accumulation detectors — `app/detect/alerts.py`
- [x] Test suite — 38 tests, all green
- [x] Engine walkthrough notebook — `notebooks/`
- [ ] FastAPI routes implementing the frozen contract
- [ ] Postgres / SQLAlchemy models; curve grid caching, invalidated on new dose events
- [ ] Registry verification — every parameter cited, `verified: true`
- [ ] FAERS extraction — **begins week 2, not week 4**

Punya — client and ingredient decomposition

Owns client/ and the openFDA integration.

- [ ] React + TypeScript + Recharts skeleton, deployed to Vercel in week 1
- [ ] Mock API server returning contract-shaped fixtures
- [ ] **Today** screen — logging in under 30 seconds. Build this first

- [ ] **Curves** screen with threshold lines overlaid
- [ ] **Alerts** screen, each alert expandable to its contributing doses
- [ ] openFDA drug label API → ingredient decomposition
- [ ] Alias normalization layer, plus manual-entry fallback

The normalization layer is the sleeper task. openFDA returns ACETAMINOPHEN, Acetaminophen, APAP and outright misspellings, all pointing at the same substance. Budget a week and it still will not be enough. The manual-entry fallback is what keeps the demo working when normalization misses — an unmatched ingredient must return nothing rather than guess, because a wrong guess silently mis-attributes a dose.

Schedule

Week	Anish	Punya	Gate
1 Sep 6–13	Routes and DB against the frozen contract	Deployed skeleton; Today screen on mocks	Contract frozen; both deployed
2 Sep 14–20	Registry verification; **FAERS extraction starts**	openFDA decomposition and normalization	Registry cited
3 Sep 21–27	FAERS backtest: sensitivity, specificity, n	Curves and Alerts against the real API	**Feature freeze Sep 27**
4 Sep 28–30	Bug fixes only	Bug fixes only	Clean-session demo works

What gets cut, in this order

Decided now, so nobody has to argue about it at 2am on 28 September.

1	**Caregiver dashboard**	already cut
2	**Pattern / correlation engine**	already cut — do not reopen
3	**Plain-English symptom input**	already cut
4	**Automated FAERS pipeline**	hand-curate ~50 cases instead
5	**Registry breadth**	21 substances down to 12, all cited
6	**Renal adjustment UI**	keep the math, ship without the profile screen

The math is not on this list. If the engine is unfinished, the project has nothing to show. Cut features, not the model.

Honest risks

The registry is the critical path, not the math

All 21 entries are currently flagged `verified: false`. Those are plausible numbers, not sourced ones. Twelve cited substances are worth more than twenty-one uncited ones, and verification is slow, unglamorous,

and cannot be parallelized away. The flag stays until a human has opened the source.

FAERS is the piece most likely to consume week 3 and return nothing

Drug names appear as brands, generics and misspellings; doses are frequently missing; the usable case count will be far below the raw report count. A measured 68% on 50 hand-verified cases beats a broken pipeline. Keep the descope path live from the start rather than reaching for it in the last week.

Whatever number we report carries its limitations in the same sentence. FAERS is voluntary and self-reported and has no denominator, so no true incidence can be computed from it. A claimed 97% would signal that we misunderstood the dataset.

Being mistaken for medical advice

One-compartment model, population-average parameters. It can deviate substantially for anyone with atypical metabolism. The framing has to be identical in the app, in the write-up, and in how we describe it out loud: we inform and route to care. We do not diagnose and we do not prescribe. Anything moving toward real distribution needs regulatory review, since software that models drug concentrations for a specific person and issues safety warnings sits near the FDA's clinical decision support boundary.

Merge hygiene

- `main` is protected; everything lands through pull requests.
- CI runs `pytest` on every push. A red build blocks the merge.
- Anish does not edit `client/`. Punya does not edit `app/pk/`.
- Shared files — the API contract and the registry JSON — change only in single-purpose pull requests.

Repository: github.com/punyagandhi/Pharmacon · 38 tests passing at time of writing. Not a medical device; educational and demonstrative use only.